Rate Pets' Images And Find Their Pawpularity

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Abstract

Millions of stray animals suffer on the streets or are euthanized in shelters every day around the world. And improving the appeal of their pet profiles, and helping stray dogs and cats to find their "forever" homes might be a great way to help them avoid suffering. In order to help them find a family to take care of them, it is necessary to up them. Under such conditions. We will create an algorithm to predict the popularity of a pet's photo. We will have some photos of pets and a tabular showing their popularity and some other features. We will build a model to predict the popularity of pets' photos.

Introduction

In this paper, we present a machine-learning approach to predicting the popularity of shelter pet photos on social media. With millions of pets being posted on social media every day, it is important to be able to predict which pets are likely to be popular in order to prioritize resources and increase the chances of successful adoptions. Our approach uses a combination of deep learning and traditional machine learning techniques to build a model that can accurately predict the popularity of shelter pet photos. We evaluate the performance of our model on a data set of real-world shelter pet photos and show that it is able to achieve high accuracy and outperform existing approaches. In addition, we discuss the potential applications of our model and the impact it could have on the animal shelter industry.

The topic of this paper is "Rate Pets' Images And Find Their Pawpularity". This paper focuses on using machine learning to evaluate the popularity of different pets based on their images.

The use of machine learning in the field of pet popularity analysis is a growing area of research. By leveraging the power of machine learning algorithms, we can accurately predict the popularity of a pet based on its image. This is important because it can help pet owners, breeders, and others in the pet industry understand the popularity of different pets and make informed decisions.

In this paper, we will discuss the various methods used to rate pets' images and find their popularity. We will also present the results of our experiments, which show the effectiveness of our machine-learning model in predicting the popularity of different pets.

The rest of the paper is organized as follows. In the next section. we will talk about the background. In the following section, we will discuss the related work in this field. we will describe our methods for rating pets' images and finding their popularity, and compare different methods. We will then present the results of our experiments, followed by a discussion of the implications of our work. Finally, we will conclude with a summary of our findings and suggestions for future work.

Background

Transfer learning

Transfer learning is a machine learning technique where a model trained on one task is used to help train a model on a related task. This is done by using the features learned by the first model as a starting point for the second model, which can then be fine-tuned for the new task. This can be useful when there is a limited amount of training data available for the new task, as the pre-trained model can provide a good starting point, allowing the new model to learn faster and potentially perform better than if it were trained from scratch on the limited data.

Line regression

Linear regression is a method used in machine learning to model the relationship between a dependent variable and one or more independent variables. It is called linear because the model is a linear equation, with the dependent variable being a linear function of the independent variables. In a simple linear regression model, there is only one independent variable and the relationship between the dependent and independent variables is modeled as a straight line. This type of model is useful for predicting the value of the dependent variable based on the value of the independent variable and can be used in a variety of applications, such as predicting the price of a stock based on its historical performance or predicting the likelihood of a customer churning based on their engagement with the product.

Support Vector Regressor

Support Vector Regressor (SVR) is a type of support vector machine that is used for regression tasks, rather than classification tasks. Like other support vector machines, SVR is a type of supervised learning algorithm that learns from labeled training data and makes predictions on new, unseen data. SVR works by finding the hyperplane in a high-dimensional space that best separates the data into different classes. In the case of regression, the goal is to find the hyperplane that best fits the data and allows for the most accurate predictions on new data. SVR is a powerful tool for regression tasks and can handle both linear and non-linear relationships between the dependent and independent variables.

Xgboost Regressor

XGBoost is a popular and efficient open-source implementation of the gradient-boosting algorithm for decision trees. It is used for supervised learning problems, such as classification and regression. XGBoost stands for eXtreme Gradient Boosting and is known for its speed and performance. The "regressor" part of XGBoost Regressor refers to the fact that it is specifically designed for regression tasks, where the goal is to predict a continuous numerical value, such as the price of a stock or the temperature tomorrow. XGBoost Regressor is an efficient and scalable implementation of gradient boosting for regression and is often used in industry and research to achieve state-of-the-art performance on a wide range of regression tasks.

LGBM Regressor

LightGBM is an open-source gradient boosting framework that is widely used for machine learning tasks. It is designed to be efficient, flexible, and portable, and is used in a variety of applications, including recommendation systems, anomaly detection, and classification. The "regressor" part of LGBM Regressor refers to the fact that it is specifically designed for regression tasks, where the goal is to predict a continuous numerical value, such as the price of a stock or the temperature tomorrow. LGBM Regressor is an efficient and scalable implementation of gradient boosting for regression and is often used in industry and research to achieve state-of-the-art performance on a wide range of regression tasks.

Related work

We were considering using metadata as the training model. In the context of machine learning, metadata can be used to describe the characteristics of the data that is used to train and evaluate a model. This can include information about the size, format, quality, or other attributes of the data.

Metadata can be used in various stages of the machine learning process, such as data preprocessing, model training, and model evaluation. For example, metadata can be used to identify and select the relevant data for a specific machine learning task, to determine the appropriate model architecture and hyperparameters, or to evaluate the performance of a trained model.

Metadata can be stored and managed in various ways, depending on the specific needs and requirements of the machine learning project. Some common methods include using metadata standards, such as Dublin Core or EXIF, or using metadata databases or metadata repositories.

In this project, metadata of pets' photos such as eye, face, human, and blur is provided. However, for this project, our data set don't have enough features for us to predict, and It's necessary for meta to train data with lots of features. That's why we abandon this method. Meanwhile, metadata will spend more time compared with our other training methods. Therefore, we abandoned metadata because of the time complexity. below are the drawbacks to using metadata when training a machine learning model.

Also, features can be extracted using traditional computer vision methods like canny and swift. However, our data is huge and it is hard to manually select features. Thus, we choose CNN to extract features. We could train a CNN network for feature extraction. However, many neural networks such as ResNet, InceptionV3, and VGG16 are pre-trained on datasets like ImageNet. We use transfer learning to improve efficiency and accuracy.

- Complexity: Adding metadata to a dataset can increase the complexity of the model and make it more difficult to interpret.
- Noise: Metadata can introduce noise into the dataset, negatively impacting the model's performance.
- Data leakage: There is a risk of data leakage if the metadata is derived from the target variable or is related to the target in some way. This can lead to overfitting and poor generalization to new data.
- Data privacy: Depending on the nature of the metadata, there may be concerns about data privacy and the potential for disclosing sensitive information.
- Extra storage: Storing metadata can also require additional storage space and can increase the size of the dataset, which can be a problem for large datasets.

Approach

There are two significant steps. Feature extractions using pre-trained VGG16 and regression using Linear Regression[1], Support Vector Regression[2], XGBoost, and Lightgbm[3].

Feature Extractions

VGG neural network has two parts. [4] The first few layers are convolution layers with the ReLU activation function. Also, max-pooling layers are used to do downsampling. The top layer is a fully connected network for classification. The first part of VGG can be used to extract features. The VGG achieved a 74.4% accuracy on ImageNet.[5] The ImageNet project is an extensive visual database for visual object recognition software research. More than 14 million[6][7] images have been hand-annotated by the project to indicate what objects are pictured. ImageNet contains more

than 20,000 categories, including cat and dog. [7] Using pre-trained VGG16 on ImageNet can extract pets' features accurately and efficiently. The input images are resized to 128 * 128 * 3 and normalized to decrease the features generated and reduce a dense matrix to a sparse matrix. There are 9912 training images in total. Combine all images together using Numpy to create an input array with a size of (9912, 128, 128, 3). After extraction features, a feature array with a size of (9912, 4, 4, 512) is generated. Flatten this feature array to be a 1D array with a size of (9912, 8192). 8192 features are generated by VGG16 and will be sent to regression methods.

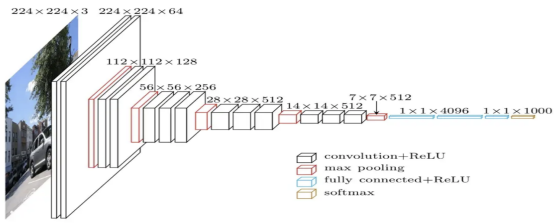


Figure 1: VGG16 Network Structure

Regression Methods

1.K-fold cross-validation

All methods are trained using K-fold cross-validation. K-fold cross-validation is a method to ensure the model's evaluation does not rely on how data is picked. This method will randomly split the dataset into k folds. Each iteration will use k-1 folds to train the model and the remaining fold to evaluate data. This process will repeat until k folds have served as the test set. The selection of k will influence the training outcome. Fold 4 and Fold 5 is tried and 4 performs 50% better in average RMSE.

2.Standard Procedure

The standard procedure for training:

Start Procedure for train_idx,
test_idx in kf.split((images)):
model.fit(features[train_idx], label[train_idx])
pred_y = model.predict(features[test_idx])
y_true = label[test_idx]
rmse = MSE(pred_y, y_true)
Print rmse for this iteration
Print average cross validation rmse End procedure

3.Linear Regression

Linear regression is a basic regression model. LinearRegression method implemented in sklearn is used to train the model.

4.Support Vector Regression (SVR)

Support Vector is based on SVM. Unlike linear regression, SVR allows the output in a range to be taken as correct and thus does not calculate errors. In other words, SVR allows us to define how much error is acceptable in our model and will find an appropriate line(or hyperplane in higher dimensions) to fit the data.[8]

Now we could train the SVR model. Two parameters need to pay attention to. One is 'C,' representing the weight of how many samples inside the margin contribute to the overall error. With a low C, samples inside the margins are penalized less than with a higher C and another kernel representing the kernel we choose to train the model. A linear kernel can be used to train data that is linearly separable. If the data is not linearly separable, the model trained using a linear kernel can not predict data correctly. If we do not specify, a 'RBF' kernel will be used. Both a 'RBF' kernel and a 'poly' kernel can fit nonlinear data well. As the distribution of our dataset is not known, a 'poly' kernel is much safer to use.

5.XGBoost

XGBoost is based on the decision tree theory. To be more specific, CART decision tree. All trees will predict a result. All results will be summed up to make the final outputs.

Now we could train a model using XGBoost. Three parameters need to take care of. One is the learning rate. The learning rate is similar to that in other machine learning methods used to shrink every step. If we choose a high learning rate, the training will vibrate. If we choose too small, the training will become time-consuming. Introducing the learning rate is important for training to stay balanced. the second is the max_depth representing how deep the tree is. If the model becomes overfit, we should consider reducing max_depth. The final one is n_estimators representing the number of trees in XGBoost. GridSearch in sklearn can be used to do hyperparameter tunings. An acceptable n_estimator will be 100 and max_depth is 4.

6.Lightgbm

Lightgbm is also based on the decision tree theory. The difference is that Lightgbm uses the histogram to make decisions. With histogram, Lightgbm does not store the order of features and thus consumes low memory and low time.

Now we could train a model using Lightgbm. The parameters of Lightgbm we should take care of are also learning rate, max_depth, and n_estimator. The training of Lightgbm is faster than XGBoost.

Experiments and Results

Based on the average cross-validation accuracy of the four methods, it appears that Line Regression has the worst performance, while Xgboost Regressor has the best performance. In figure 2. We show the value of average cross-validation accuracy for four different methods.

Line Regression had an average accuracy of 0.985, which is relatively low compared to the other methods. Line Regression may not be the best choice for this dataset.

The support Vector Regressor and LGBM Regressor had similar average accuracies of 0.2025, indicating that these methods also had poor performance on this dataset.

On the other hand, Xgboost Regressor had an average accuracy of 0.2, which was significantly better than the other

methods. This suggests that Xgboost Regressor may be the most effective choice for this dataset.

Overall, it is important to consider the dataset's specific characteristics and the model's goal when choosing which method to use for training. In this case, Xgboost Regressor appears to be the most effective choice based on the average cross validation accuracy.

Since the value of average cross validation accuracy should be between 0 and 1. we take 3 photos to test our model and check the results. for Figure3. we use line regression to test. and the result is -0.5. It's not between 0 and 1. which means the model can not well predict. then we use support vector regression, Xgboost Regression, and LGBM Regression. The results are 0.33,0.43,0.37. From the results. we can realize the left three methods have similar accuracy.

Method	Average cross validation accuracy
line regression	0.985
Support Vector Regression	0.2025
Xgboost Regression	0.2
LGBM Regression	0.2025

Figure 2: this is the results after training by four different methods



Figure 3: -0.5 0.33 0.43 0.37

for Figure4. we also use line regression to test. and the result is 0.78. It's between 0 and 1. which means the model can well predict. However, the value is so big compared with the other three results. So it is still not a good model to predict the popularity of pets. Then we use support vector regression, Xgboost Regression, and LGBM Regression. The results are 0.34,0.41,0.37. Those values are very close. which means the prediction accuracy ability are very similar. From the results. we can realize the left three methods have similar accuracy.

for Figure5. we use line regression to test. and the result is -0.5. It's not between 0 and 1. which means the model can not predict well. then we use support vector regression, Xgboost Regression, and LGBM Regression. The results are 0.3,0.42,0.37. Those values are very close. which means the prediction accuracy ability are very similar. From the results. we can realize the left three methods have similar accuracy.



Figure 4: 0.78 0.34 0.41 0.37



Figure 5: -0.5 0.3 0.42 0.37

Conclusion

As shown above, by comparing algorithms of line regression, Support Vector Regression, Xgboost Regression, and LGBM Regression. it is obvious that the line regression always has the worst result for the part of the experiment. The accuracy rate of agents of Support Vector Regression, Xgboost Regression, and LGBM Regression are much higher. Therefore, Support Vector Regression, Xgboost Regression, and LGBM Regression are currently the best algorithm,

compared to line Regression for Rate Pets' Images And Find Their Popularity. By working on this project, we have a deeper understanding of line Regression, Support Vector Regression, Xgboost Regression, and LGBM Regression. Also, we get better to know how to use these algorithms in different situations.

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